

Dietary flaxseed alters tumor biological markers in postmenopausal breast cancer.



PURPOSE: Flaxseed, the richest source of mammalian lignan precursors, has previously been shown to reduce the growth of tumors in rats. This study examined, in a randomized double-blind placebo-controlled clinical trial, the effects of dietary flaxseed on tumor biological markers and urinary lignan excretion in postmenopausal patients with newly diagnosed breast cancer. EXPERIMENTAL DESIGN: Patients were randomized to daily intake of either a 25 g flaxseed-containing muffin (n = 19) or a control (placebo) muffin (n = 13). At the time of diagnosis and again at definitive surgery, tumor tissue was analyzed for the rate of tumor cell proliferation (Ki-67 labeling index, primary end point), apoptosis, c-erbB2 expression, and estrogen and progesterone receptor levels. Twenty-four-hour urine samples were analyzed for lignans, and 3-day diet records were evaluated for macronutrient and caloric intake. Mean treatment times were 39 and 32 days in the placebo and flaxseed groups, respectively. RESULTS: Reductions in Ki-67 labeling index (34.2%; $P = 0.001$) and in c-erbB2 expression (71.0%; $P = 0.003$) and an increase in apoptosis (30.7%; $P = 0.007$) were observed in the flaxseed, but not in the placebo group. No significant differences in caloric and macronutrient intake were seen between groups and between pre- and posttreatment periods. A significant increase in mean urinary lignan excretion was observed in the flaxseed group (1,300%; $P < 0.01$) compared with placebo controls. The total intake of flaxseed was correlated with changes in c-erbB2 score ($r = -0.373$; $P = 0.036$) and apoptotic index ($r = 0.495$; $P < 0.004$). CONCLUSION: Dietary flaxseed has the potential to reduce tumor growth in patients with breast cancer.